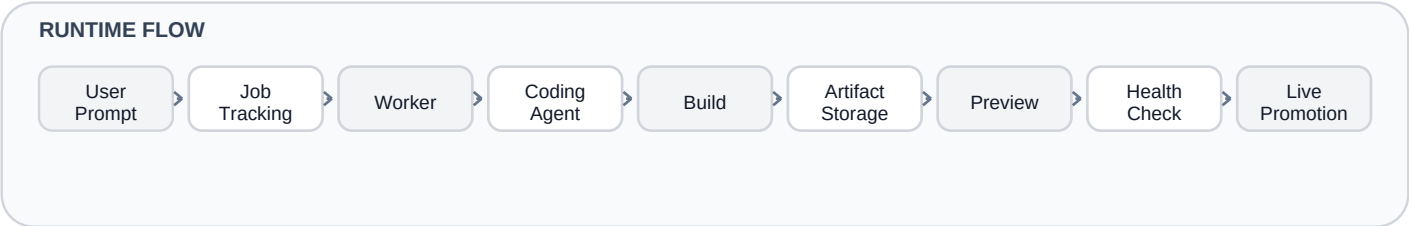


Softbox — AI Runtime for Safe App Modification and Promotion

Selected work sample for AI engineering, orchestration, and deployment workflow projects

PROJECT SUMMARY

Softbox is a runtime for AI-modified applications. Instead of letting an agent change code and deploy it directly, it routes each prompt through a controlled workflow: rewrite, build, preview, validate, then promote. The result is a practical system for AI-driven app changes with deployment safety built in. It matters because the engineering problem is not generating code alone, but turning generated code into a release process that is observable, testable, and safe to promote.



PROBLEM

AI can modify application code quickly, but direct handoff from model output to production is risky. The core problem is creating a workflow where generated changes can be evaluated without exposing the live app to unstable code paths.

SOLUTION

Softbox separates the system into a stable outer shell and a mutable inner app. A prompt creates a tracked job, a worker sends the selected app to a coding agent, the agent rewrites source, and the worker builds a new immutable artifact. That artifact is mounted in preview inside the shell, health-checked, and only then promoted live.

TECHNICAL HIGHLIGHTS

- AI-assisted code rewrite pipeline
- Job orchestration and state tracking
- Immutable build and version workflow
- Preview environment before promotion
- Deployment gating via health checks
- Backend integration with Convex and Cloudflare R2

WHY THIS IS RELEVANT FOR CLIENT WORK

This project demonstrates real AI application engineering rather than model demos alone. It maps directly to backend implementation, workflow automation, API and system integration, and production-minded release logic for AI-powered products.

MY CONTRIBUTION

I designed and built the runtime workflow end to end. I implemented the orchestration between prompt submission, job tracking, worker execution, agent-driven code changes, artifact creation, preview mounting, and promotion. I also handled backend and systems concerns including versioning, artifact flow, validation rules, and deployment safety.

STACK

AI agent-based code modification

Convex

Cloudflare R2

Backend/runtime orchestration